EART27201— Sedimentary Rocks and Fossils
Fossil Practicals — Graptolites and Conclusion

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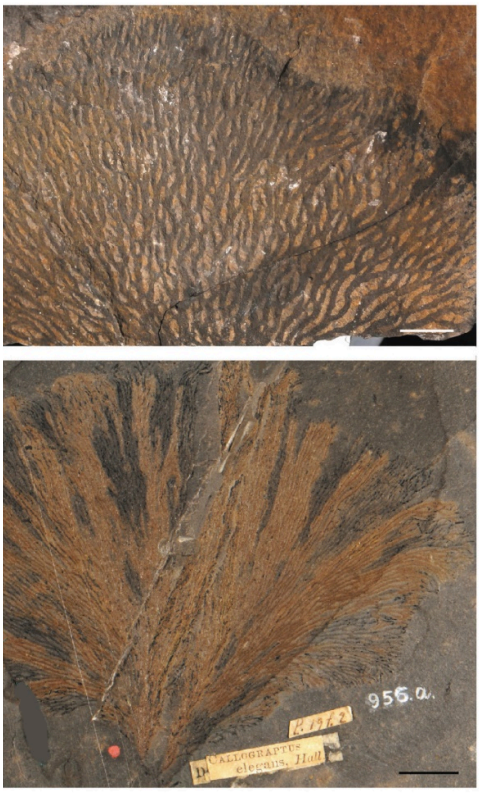
Often there will be QR codes that link to online resources. If you are accessing the page via e.g. a laptop (so resolving the QR code is challenging), you can find a list of all the links at this URL:

https://www.russellgarwood.co.uk/EART27201.html

Under each QR code in this booklet is a keyword in italics - click the associated link on the page above, and it will take you to the correct target.

Well hello there. This week marks the end of us looking at big stuff. Before we go onto small stuff next week, let's finish up with some more practical exercises!

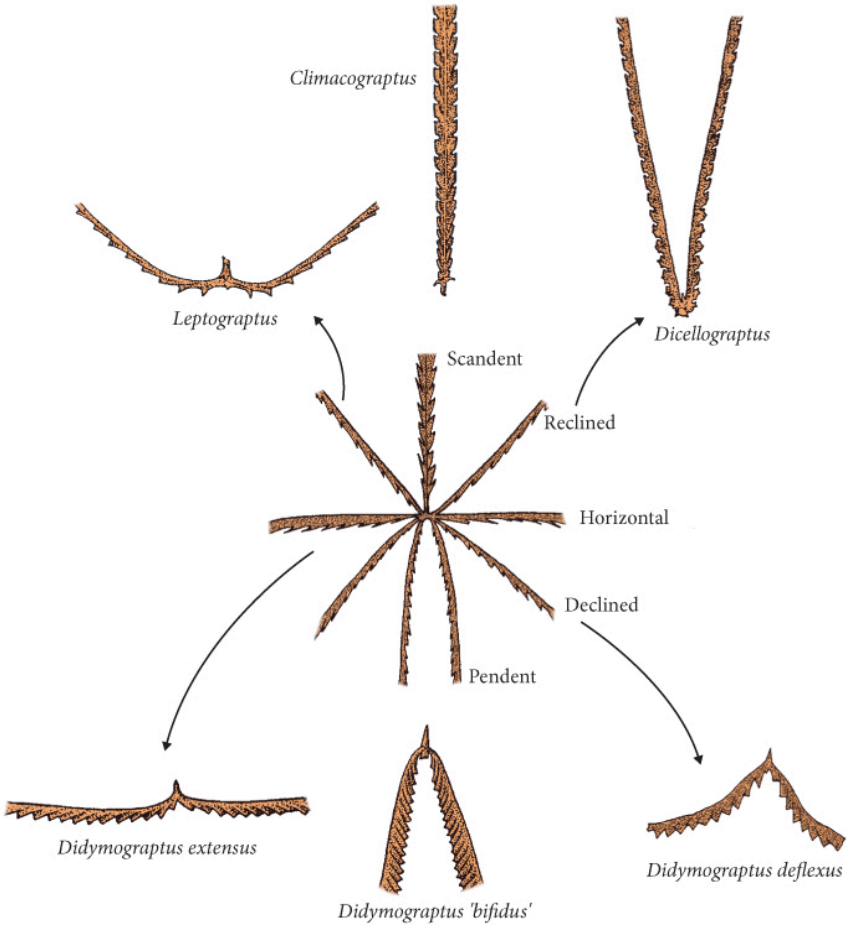
Graptolites



The origins of the graptolites lies in the Cambrian period; the group probably evolved from a solitary, worm-like animal which had arms and tentacles. These early representatives of the group might have been pseudocolonial filter-feeders, found in clumps on the seafloor. As the Cambrian progressed clumps became more erect structures, and by the end of the Cambrian we see a range of dendroid forms. These are multibranched colonies resembling small shrubs, attached to the substrate by holdfasts, some of which are shown left (scale bars 10mm).

It was during the late Cambrian and Early Ordovician that some of these dendroid graptolites moved from the benthic communities of the time to a planktonic mode of life – and from dendroid graptolites, graptoloid graptolites evolved! There is then an evolutionary trend towards fewer branches (from eight down to two, then one), and the group evolved through a large number of forms, graptoloids being dominant from the Ordovician.

This trend included a change with the position of the stipes, otherwise known as their attitude, which can be defined as pendent, declined, horizontal reclined or scandent. These are shown labelled on the right. Roughly speaking, from the bottom to the top of this diagram we're tracking the evolution of the attitude of this group through the Ordovician and into the Silurian Period.



This rather famous evolutionary sequence is shown in a diagram at the top of the next page. It kicks off in the early Ordovician with *Tetragraptus*, typically comprising four stipes. Coincident with this a second genus, *Didymograptus* appeared, which had two stipes (it was biramous), which could be pendent, horizontal, or reclined in attitude. As time passed, increasingly fossils display horizontal arrangement of their stipes, followed by the evolution of more reclined attitudes. These sometimes had curved

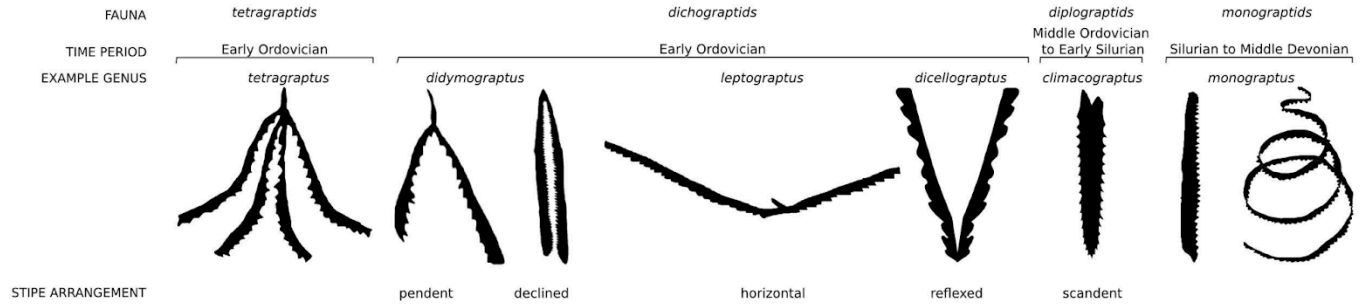


Image Source: University of Cambridge, 1A Collections. https://wserv3.esc.cam.ac.uk/p1acollections/items/show/114.

or even coiled stipes. As this sequence unfolded, thecae became more complex in shape. Once we hit the middle Ordovician, and into the Silurian, we find fossils of scandent graptoloids - this is where the thecae are found on the outside of the colony. After this, forms appear where a stipe has been lost – these are known as uniserial scandent graptoloids, such as we see in the genus *Monograptus*. Monograptids such as this dominated graptolite faunas in the Silurian, and developed a huge variety of forms that can be identified via the shape and arrangement of the thecae. The last graptoloids are found in Early Devonian rocks. We don't, however, know for sure what drove this evolutionary sequence. Was the simpler stipe configuration more hydrodynamically stable thus better adapted to turbulence? Did it assist graptolites when moving through the water column feeding? Did it prevent the interference between adjacent stipes and their thecae, thus improving efficiency? All have been suggested.

Graptolite task 1: In front of you are examples of two fossil graptolites (53 and 2). Where on the evolutionary sequence described does each fossil fall? Base your answer on the attitude of the stipes, and make sure to see if you can spot the individual thecae in your fossils (this can be quite challenging as sometimes they are small!). How old is the rock each is found in likely to be?

Graptolite task 2: What environment of deposition is the rock likely to represent - the fossils will provide an answer, but you can also use sedimentary features to suggest an answer if you know these.

Benthic Assemblages

In today's lecture we learned that there are biofaces associated with depth on clastic shelves, and that these are reflected in benthic assemblages. Now let's meet them in a bit more detail.

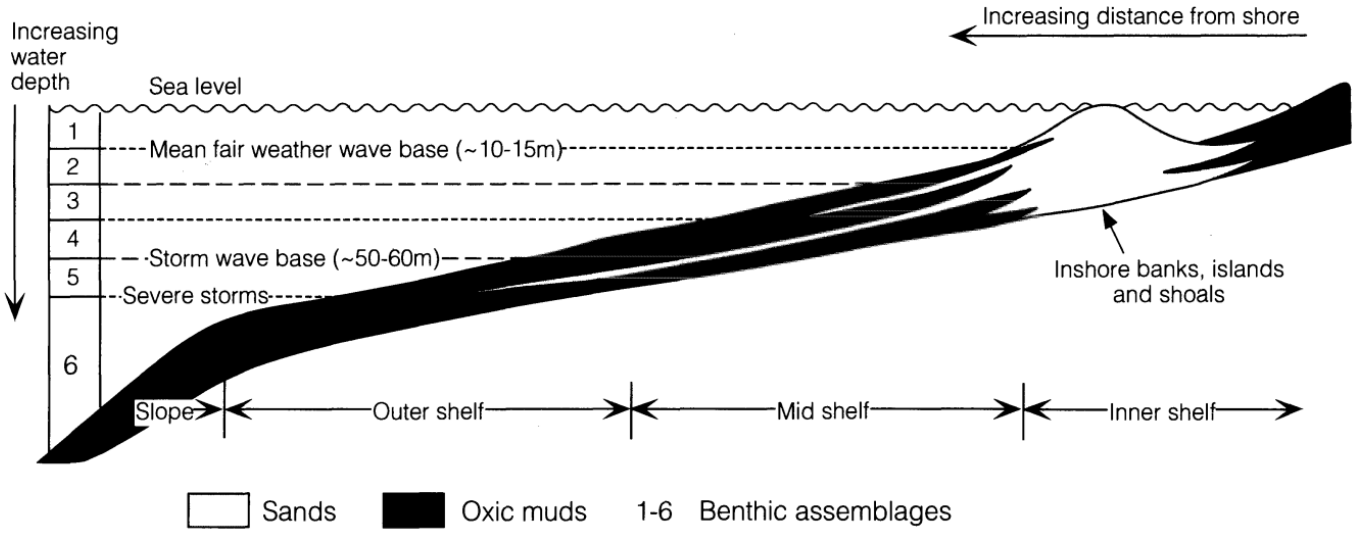
For this practical, we will use a website. You can find this by visiting *Conclusion* on from the links website, or following the QR code to the right. This site provides insights into how holidays with geologists often turn out. On the page are a series of fossils (both trace and body fossils), and a few sedimentary structures, of the kind you might expect to see at two different outcrops in the USA if you were, say, on a walking holiday in New York State. Have a look at these fossils! On the following pages you will find a series of exercises that draw together lots of bits and pieces that you have learned about macrofossils, and include some associated information on trace fossils and sediments. The information on the 3D models has been removed, so you'll have to try and identify them based on the models, and build up some inferences from there.

Link *Conclusion*

To help you out, are some tables (below) that provide an overview of our benthic assemblages. These are taken from the following book:

Brenchley, P.J., Brenchley, P. and Harper, D., 1998. Palaeoecology: Ecosystems, environments and evolution. CRC Press.

Which is also a great place to learn more about them! This diagram, which is also from that book, reminds us of their rough depth:



Brackish Lagoonal Environments

Context	Commonly associated with terrestrial, barrier bar, estuarine, marsh or shallow marine shoreface facies.
Lithology	Typically shales though siltstone, sandstones and less commonly coarser grained sediments may be introduced by tidal or storm processes in flood-tidal deltas and washover fans.
Taphonomy	Dispersed in situ or disturbed neighbourhood assemblages, typically as thin shell concentrations. Tidal and storm processes may introduce allochthonous assemblages from shoreface and inner shelf environments.
Taxonomy/Diversity	In situ faunas typically of low diversity; usually ostracodes, bivalves and gastropods.
Abundance	Variable, commonly low in thin shell accumulations, but high in thick, typically monospecific, shell beds.
Ichnology	Imperfectly described; low diversity assemblages of the Psilonichnus, Glossifungites, Cruziana and Zoophycos ichnofacies may each occur depending on energy and salinity levels, grain size, substrate consistency and specific depositional environment within a lagoon.

Shoreface Environments (BA1) 0-5m

Context	Nearshore and associated beach facies, distinguished from inner shelf deposits by generally having only rare mudstones interbedded with sandstones. Shoreface facies commonly cap upward-coarsening shallow marine cycles.
Lithology	Typically sandstones with locally developed mudstones. Sandstones exhibit planar- or cross-stratification formed by wave, tidal or wind-forced currents, or hummocky cross-stratification formed by oscillatory or combined flow.
Taphonomy	Allochthonous shell concentrations that may occur as coquinas, on bedding planes or cross-stratified foresets.
Taxonomy/Diversity	Palaeozoic assemblages characterized mainly by brachiopods with more rarely bivalves and gastropods. Mesozoic/Cenozoic assemblages characterized by varied bivalves (including deep burrowers and attached forms) and gastropods. Taxa affected by substrate mobility (e.g. Corals) are rare. In situ faunas rare. Diversity, though still generally low, can be elevated by introduction of allochthonous faunas by storm, tidal and wave activity.
Abundance	Generally low in Palaeozoic rocks, but Cenozoic strata typically possess more abundant remains. In more subdued shoreface environments shells may be relatively common in coquinas.
Ichnology	Typically Skolithos, more rarely Psilonichnus and Cruziana ichnofacies depending on substrate mobility and subtle gradients in hydrologic sedimentologic and ecologic parameters. Diversity typically low.

Inner (shallow) Shelf Environments (BA2) 5-15m

Context	Facies subject to frequent storm waves and tidal processes, commonly occurring below shoreface sandstones and above mid-shelf sequences in coarsening-upward cycles.
Lithology	Sandstones dominate, but may occur in equal abundance with interbedded shales. Storm-influenced facies exhibit hummocky cross-stratification; tidal facies may exhibit herringbone cross-stratification, sigmoidal cross-stratification or mud drapes.
Taphonomy	Allochthonous or disturbed neighbourhood assemblages at the base of storm sandstones, typically as coquinas. Tidally influenced sandstones commonly exhibit transported assemblages or shell lags distributed along foresets. Shales possess disarticulated and dispersed shells and bedding plane assemblages. In situ faunas rare or absent.
Taxonomy/Diversity	Palaeozoic strata dominated by moderately but variably diverse assemblages of brachiopods, particularly orthoids and strophomenoids: Mesozoic/Cenozoic strata typically possess variable infaunal and epifaunal bivalves. Representatives of other benthic macroinvertebrates may occur, sometimes in abundance.
Abundance	Generally moderate, but high in coquinas. Faunas can, however, be generally sparse where a freshwater influence is established, as, for example, in inner shelf environments in front of deltas.
Ichnology	Typically moderate to high diversity assemblages of the Cruziana and/or Skolithos ichnofacies. Mudstones commonly intensely bioturbated and possess trace fossils typical of the Cruziana ichnofacies: associated sandstones contain representatives of the Skolithos ichnofacies providing substrate mobility does not result in their destruction.

Middle Shelf Environments (BA3 and BA4) 15-30m

Context	Commonly form portions of upward-coarsening sequences, but may also form part of a predominantly mudstone sequence. Below fair weather wave base, above storm wave base.
Lithology	Mudstones, typically intensely bioturbated, and interbedded sandstone tempestites with hummocky cross-stratification are common.
Taphonomy	Mudstones may contain in situ clumped or dispersed shells that are still commonly articulated, or as thin stringers or locally reworked disturbed neighbourhood assemblages. Sandstones commonly contain allochthonous lag assemblages or coquinas that are taxonomically similar or dissimilar to those assemblages within the associated mudstones.

Taxonomy/Diversity	Palaeozoic strata possess moderate to high diversity assemblages dominated by brachiopods represented by several orders. Crinoids, bryozoans, corals, trilobites and gastropods also common. Mesozoic/Cenozoic strata dominated by varied infaunal and epifaunal bivalve assemblages with gastropods, echinoderms, bryozoans and less commonly brachiopods also present.
Abundance	Generally high, decreasing with depth.
Ichnology	Similar to inner shelf environments with high diversity assemblages typical of the Arenicolites ichnofacies in sandstones, typically as opportunists following storm activity, and intense bioturbation by members of the Cruziana ichnofacies in associated mudstones.

Outer (deep) shelf environments (BA5) 30-60m

Context	Mudrock-dominated sequences in association with shelf or slope/basin environments. Mainly below effective storm wave base, though extreme storms have disturbed shells.
Lithology	Bioturbated mudstones with uncommon, and where present typically thin distal siltstone or sandstone tempestites.
Taphonomy	In situ dispersed, articulated faunas predominate. Thin stringers and bedding plane assemblages also common: disturbed neighbourhood assemblages may occur. Distal tempestites may contain allochthonous shells, though not commonly.
Taxonomy/Diversity	Palaeozoic strata dominated by brachiopods and trilobites: Mesozoic/Cenozoic strata by bivalves, particularly infaunal species - nuculoids are very common. Other groups also occur but in lower numbers. Pelagic forms eg. graptolites (Palaeozoic), ammonites, belemnites (Mesozoic) also commonly present. Diversity variable depending on parameters such as nutrients, oxygen levels, etc., but can be high, then typically markedly decreases at the shelf edge.
Abundance	Typically low.
Ichnology	Discrete ichnotaxa commonly difficult to recognise as a result of intense bioturbation. Representatives of Cruziana, Zoophycos or rarely Nereites ichnofacies may predominate depending on specific environmental conditions. Bioturbation intense though diversity commonly low to moderate.

Slope and Basin (BA6 and below) 60-200m

Context	Mudrock-dominated sequences with associated sediment gravity flows particularly, but not exclusively, turbidites. Submarine canyons and associated fans commonly occur in association with slope and continental rise sequences. Contourites also common on slope and continental rise environments.
Lithology	Pelagic and hemipelagic mudstones, turbiditic sandstones, and siltstones. Canyons and fans typically possess coarser grained lithofacies.
Taphonomy	Autochthonous faunas, typically dispersed and articulated, in pelagic/hemipelagic mudstones. Allochthonous faunas introduced particularly by turbidity currents.
Taxonomy/Diversity	Dominated by pelagic macroinvertebrates (eg. Palaeozoic - trilobites, graptolites, nautiloids; Mesozoic - ammonites, belemnites). In situ forms include Palaeozoic trilobites, hyolithids, etc. and Mesozoic/Cenozoic bivalves, gastropods and crinoids. Other groups may occur but are uncommon. Diversity is low.
Abundance	Very low, but can be higher in sediment gravity flow deposits.
Ichnology	Variable, with representatives of the Skolithos, Arenicolites and/or Glossifungites ichnofacies commonly occurring in association with well-oxygenated submarine canyons and fans; Zoophycos ichnofacies with slopes with restricted circulation and low oxygen levels and the Nereites ichnofacies in more stable, classical flysch-like settings in generally quiet, but oxygenated waters.

Benthic assemblages task 1: What benthic assemblage do the fossils from locality one represent? Be as specific as you can, and justify your reasoning!

Benthic assemblages task 2: How old are the rocks cropping out at locality one?

Hint: Don't forget the first part of today's practical.

Benthic assemblages task 3: What benthic assemblage do the fossils from locality two represent? Be as specific as you can, and justify your reasoning!

Benthic assemblages task 4: What is the likely difference in water depth, roughly, at which the two rocks were deposited.

Benthic assemblages task 5: Oh my goodness, that mystery fossil! What on earth could it be? Have a guess!

Hint: It's not a graptolite! Also bear in mind that shallower environments might be expected to have sediment/fossil input from terrestrial (=land) environments.

Benthic assemblages task 6: Based on all the evidence above, are the rocks at locality one and locality two likely to represent deposits that were laid down at the same time or do they represent sediments from different time periods? If the former, you have been walking through a lateral change in depositional environment in your hike, but if the latter you could be walking up and down stratigraphy, or across a fault.

And that is it for large fossils. We're back next week with small ones!

Sources

Benton, M.J. and Harper, D.A., 2010 & 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons. (Diagrams, images, unless otherwise noted).

The Digital Atlas of Ancient Life (3D models).